DOCUMENT: PCI-DSS v4.0 Mandatory Security & Audit Specification (Q4 Audit Gate)
CLASSIFICATION: Mandatory Engineering Compliance Requirements

Requirement 1: Envelope Encryption for Cached Checkout Payloads (CONFLICTS WITH FIN-106)
- All cached merchant checkout and cardholder session payloads in Redis MUST use AES-256-GCM envelope encryption backed by AWS KMS per-tenant data keys.
- NOTE: This directly conflicts with existing backlog item FIN-106 (Plaintext Session Payload Caching in Shared Redis Cluster), which must be superseded or blocked immediately.

Requirement 2: Immutable Tamper-Evident Security Audit Trail (NEW)
- Every administrative privilege escalation, API key rotation, and SSO configuration change must emit a structured JSON audit event signed with HMAC-SHA256 to an append-only S3 Object Lock bucket.
- Retention window for cold archival storage (7 years vs 10 years) is pending Legal sign-off.